

・6分の1公式 (数学の用語ではない)

$$\int_{\alpha}^{\beta} (x-\alpha)(x-\beta) dx = -\frac{1}{6}(\beta-\alpha)^3$$

(証明)

$$\begin{aligned} & \int_{\alpha}^{\beta} (x-\alpha)(x-\beta) dx \\ &= \int_{\alpha}^{\beta} (x-\alpha)(x-\alpha+\alpha-\beta) dx \quad \leftarrow \text{この計算テクニックが大事} \\ &= \int_{\alpha}^{\beta} \{(x-\alpha)^2 + (\alpha-\beta)(x-\alpha)\} dx \\ &= \left[\frac{1}{3}(x-\alpha)^3 + (\alpha-\beta) \cdot \frac{1}{2}(x-\alpha)^2 \right]_{\alpha}^{\beta} \\ &= \frac{1}{3}(\beta-\alpha)^3 + (\alpha-\beta) \cdot \frac{1}{2}(\beta-\alpha)^2 \\ &= -\frac{1}{6}(\beta-\alpha)^3 \quad \square \end{aligned}$$

(例) 次の定積分を求めよ。

$$(1) \quad \int_{-1}^3 (x+1)(x-3) dx = -\frac{1}{6} \{3-(-1)\}^3 = -\frac{32}{3} //$$

$$(2) \quad \int_{1-\sqrt{3}}^{1+\sqrt{3}} (x^2-2x-2) dx$$

$x^2-2x-2=0$ の解は

$$x = 1 \pm \sqrt{3}$$

であるから

$$\begin{aligned} & \int_{1-\sqrt{3}}^{1+\sqrt{3}} (x^2-2x-2) dx \\ &= \int_{1-\sqrt{3}}^{1+\sqrt{3}} \{x-(1-\sqrt{3})\} \{x-(1+\sqrt{3})\} dx \\ &= -\frac{1}{6} \{ (1+\sqrt{3}) - (1-\sqrt{3}) \}^3 \\ &= -4\sqrt{3} // \end{aligned}$$

(例2) 次の定積分を求めよ。

$$\begin{aligned} (1) \quad & \int_{\alpha}^{\beta} (x-\alpha)^2 (x-\beta) dx \\ &= \int_{\alpha}^{\beta} (x-\alpha)^2 (x-\alpha+\alpha-\beta) dx \\ &= \int_{\alpha}^{\beta} \{(x-\alpha)^3 + (\alpha-\beta)(x-\alpha)^2\} dx \\ &= \left[\frac{1}{4}(x-\alpha)^4 + (\alpha-\beta) \cdot \frac{1}{3}(x-\alpha)^3 \right]_{\alpha}^{\beta} \\ &= \frac{1}{4}(\beta-\alpha)^4 + (\alpha-\beta) \cdot \frac{1}{3}(\beta-\alpha)^3 \\ &= -\frac{1}{12}(\beta-\alpha)^4 // \end{aligned}$$

$$\begin{aligned} (2) \quad & \int_{\alpha}^{\beta} (x-\alpha)^2 (x-\beta)^2 dx \\ &= \int_{\alpha}^{\beta} (x-\alpha)^2 (x-\alpha+\alpha-\beta)^2 dx \\ &= \int_{\alpha}^{\beta} (x-\alpha)^2 \{(x-\alpha)^2 + 2(\alpha-\beta)(x-\alpha) + (\alpha-\beta)^2\} dx \\ &= \int_{\alpha}^{\beta} \{(x-\alpha)^4 + 2(\alpha-\beta)(x-\alpha)^3 + (\alpha-\beta)^2(x-\alpha)^2\} dx \\ &= \left[\frac{1}{5}(x-\alpha)^5 + 2(\alpha-\beta) \cdot \frac{1}{4}(x-\alpha)^4 + (\alpha-\beta)^2 \cdot \frac{1}{3}(x-\alpha)^3 \right]_{\alpha}^{\beta} \\ &= \frac{1}{5}(\beta-\alpha)^5 + 2(\alpha-\beta) \cdot \frac{1}{4}(\beta-\alpha)^4 + (\alpha-\beta)^2 \cdot \frac{1}{3}(\beta-\alpha)^3 \\ &= \frac{1}{30}(\beta-\alpha)^5 // \end{aligned}$$